

Closed-Loop Convergence: Towards an Autonomous Network Topology Engine

Bridging Intent Drift via Automated Remediation, Digital Twins, and Semantic Knowledge Graphs

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Abstract. The Network Topology Engine (NTE) framework has successfully established a narrow-waist architecture for network state. However, the current framework operates as an open-loop system where "Intent Drift" (Δ) requires manual intervention. This report proposes an expansive, autonomous ecosystem. We introduce the *Remediation Engine* for automated correction, the *Digital Twin* for forward-simulation, and a *Semantic Network Ontology* that moves beyond topological primitives to business-aligned knowledge graphs. Furthermore, we explore *Swarm NTE* for decentralized orchestration and the role of *Agentic Reasoning Layers* in interpreting high-level architectural intent. By treating the network as a self-correcting semantic graph, we define the foundations for high-velocity, autonomous network convergence.

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1 Introduction: Beyond the Narrow Waist

The foundational works of the NTE portfolio defined the Plan Topology (G_{plan}) as the "Anchor of Reality" [0]. While this decoupled intent from vendor execution [0], it remains a *reactive* system. Telemetry-derived "Intent Drift" (Δ) is visualized but not automatically rectified [0].

This report outlines the "Integrated NetView" architecture, a multi-layered ecosystem that closes the loop between observation, reasoning, and execution.

Insight:
A truly autonomous network must not only detect drift but must possess the "Self-Healing Vector" required to converge back to the intended state.

2 The Integrated NetView Ecosystem

The NetView workbench serves as the gravitational center for the NTE repositories. Fig. 1 illustrates how these components form a coherent lifecycle.

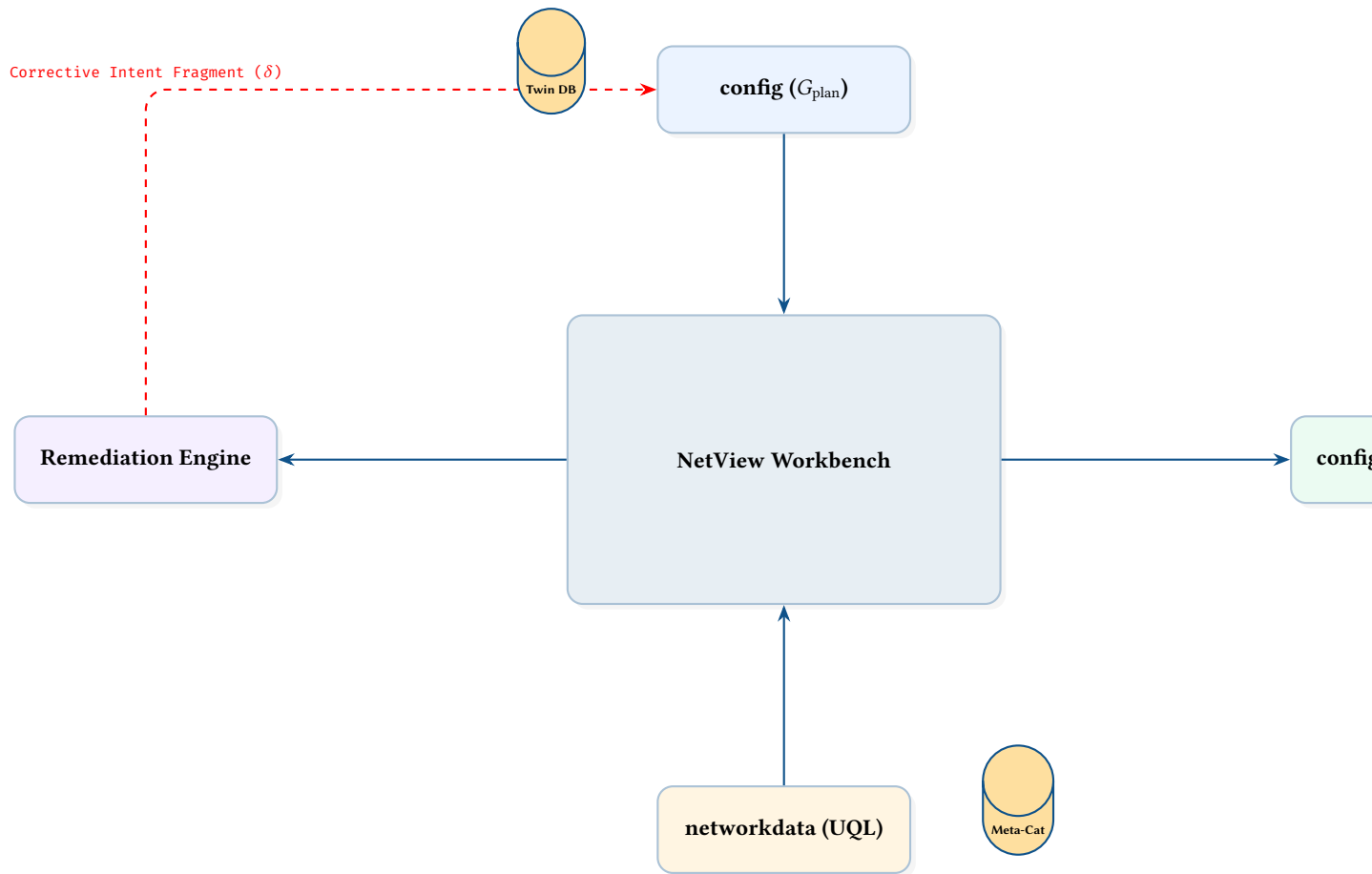


Figure 1. The Integrated NetView Ecosystem. The Remediation Engine closes the loop by injecting algebraic fragments (δ) back into the source-of-truth.

3 The Semantic Network Ontology

Traditional IBN models focus on topological primitives (VRFs, BGP peers). An expansive architecture requires a **Semantic Knowledge Graph** that links physical assets to user identity and business constraints.

Design Rule 1: Semantic Grounding

Every topological element in G_{plan} MUST be grounded in the Semantic Ontology. An "Interface" is not just a port; it is a mapping $\mathcal{M} : \text{Port} \rightarrow \text{Service} \rightarrow \text{Legal_Jurisdiction}$.

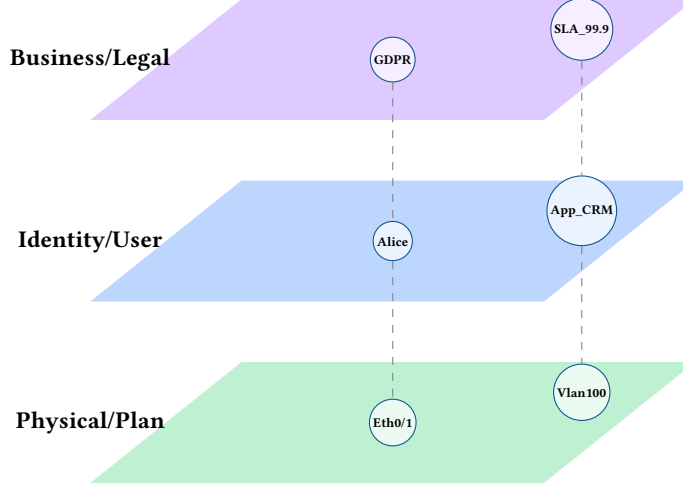


Figure 2. The Semantic Knowledge Graph. Intent is propagated from business constraints (e.g., GDPR data residency) through user identity down to the physical plan topology.

4 The Digital Twin Sandbox: Parallel Universes

A core missing piece is the ability to "Time-Travel" or simulate "What-If" scenarios. The **Digital Twin** creates parallel graph instances.

Before the Remediation Engine commits δ , the Digital Twin spawns $G_{\text{twin}} = G_{\text{plan}} \oplus \delta$ and runs a traffic-flow predictor to ensure the remediation doesn't cause cascading congestion.

Insight:
We move from "Validation by Compile" to "Validation by Simulation."

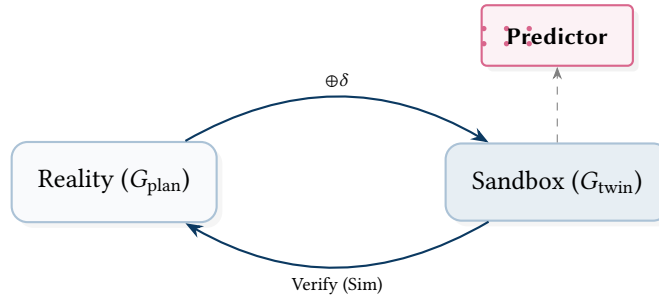


Figure 3. Digital Twin Verification. Parallel graph instances are used to test remediation stability before deployment.

5 Swarm NTE: Decentralized Orchestration

The expansive vision replaces the centralized hub with a **Swarm of NTE Nodes**. Each node manages a local subgraph but exchanges "Intent Pressure" vectors with neighbors.

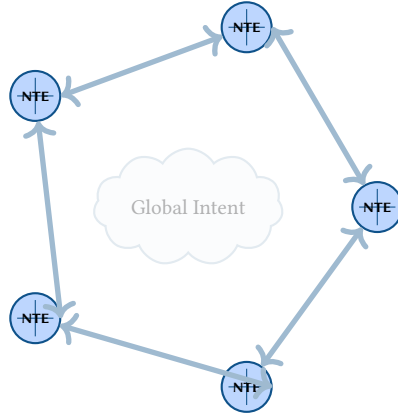


Figure 4. Swarm NTE Architecture. Decentralized nodes negotiate local subgraphs to satisfy global intent constraints.

6 The Agentic Reasoning Layer: The LLM Architect

Finally, we introduce the ****Agentic Reasoning Layer****. LLMs are not used for coding, but for ****Grounding Natural Intent****.

This layer interprets the "Whys" behind the architecture, ensuring that the Remediation Engine's δ fragments align with the high-level architectural spirit, not just the bit-level constraints.

7 Conclusion

By fusing the Semantic Ontology, Digital Twin simulations, and Decentralized Swarm logic, the NTE framework moves from a static "Narrow Waist" to a dynamic, "Living Graph." This closed-loop convergence ensures the physical network is a permanent, self-correcting reflection of architectural intent.

Insight:
The LLM acts as the translator between "Make the Finance network redundant across sites" and the specific UQL/UPL Primitives required to compose the G_{plan} .

Revision History

Version	Date	Changes
1.0.0	March 2026	Initial specification of the closed-loop autonomous framework.
1.1.0	March 2026	Expanded with Semantic Ontology, Swarm NTE, and Agentic Reasoning.